

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

1. Introduction

7 hrs · asked in **all 41 papers** · 6 syllabus sub-topics · 337 marks, 9.9 % of every mark ever set · no numerical component.

Read this first:

- **Every paper opens here.** Q1 is nearly always a two- or three-part Introduction question for 7 or 8 marks, and it is the cheapest 8 marks in the subject.
- Four questions carry the whole chapter:
 - **Turing test** **TOP 10/41** – what it is, when it is passed, what a machine needs to pass it.
 - **Applications of AI** **TOP 8/41** – almost always the tail of the Turing-test question, never a question on its own.
 - **Intelligent agents** **HOT 7/41** – define, then the types, with diagrams.
 - **PEAS** **HOT 7/41** – write the four rows for a named system.
- **The one idea everything hangs off is the four-approaches grid** (§1.1). Thinking Vs acting, humanly Vs rationally. Every *define AI* answer is that grid; the Turing test is the *acting humanly* box; the rational agent is the *acting rationally* box. Learn the grid and three separate questions collapse into one.
- Pairs the examiner reliably asks together: **Turing test** + **applications** (7 papers), **intelligent agent** + **PEAS** (6 papers), **knowledge** + **learning** (5 papers).
- **PEAS is worth memorising cold.** It is asked in 9 papers for a *named* system (taxi, medical diagnosis, self-driving car, robot in a manufacturing plant, medicine-delivery drone) and no lecture deck works a full example. Four rows, four columns, §1.4.
- Do not write history as prose. The examiner wants **year** → **event** lines, and one of those questions asks specifically which period was the “AI research failure” and why.

1.1 What Artificial Intelligence is

Define Artificial Intelligence. Describe AI and its subfields **1 2 4**

TOP 10/41 70 Bh, 69 Po, 75 Ba, 77 Ch, 78 Ba, 79 Ash, 79 Bh, 81 Ba, 77 Ch, 76 Bh · opens the Turing-test question in almost every paper, so it is never worth more than 2 marks on its own, and it is never skippable

Definitions worth quoting

- **Elaine Rich:** the study of how to make computers do things at which, at the moment, people are better.
 - Best one to write: it explains why the field keeps moving. Once machines win at a task, it stops counting as AI.
- **Kurzweil:** the art of creating machines that perform functions requiring intelligence when performed by people.
- **Minsky:** the science of making computers do things that require intelligence when done by humans.
- **Working definition:** a branch of computer science that builds machines which solve complex problems in a human-like fashion.

Intelligence, unpacked

- Ability to **understand**, to **reason**, to **learn**, to **create**.
- No machine has all four. That gap is the whole subject.

Intelligent behaviour: three task classes

- **Everyday tasks:** recognise a friend, recognise a caller's voice, translate a language, interpret a photograph, talk, cook a meal.
 - Easiest for people, **hardest for machines**. Worth saying: this is Moravec's paradox.
- **Formal tasks:** prove a theorem, do calculus, play chess, checkers, Go.
- **Expert tasks:** engineering design, medical diagnosis, financial analysis.
 - Hardest for people, **easiest for machines**, because the knowledge is explicit and narrow.

Subfields of AI

- Search and problem solving · knowledge representation and reasoning · planning · machine learning · neural networks · expert systems · natural language processing · machine vision and image processing · robotics · fuzzy logic and evolutionary computing.

The four approaches to AI (the grid everything sits on) 4 7 8

74 Bh, 76 Ba, 82 Bh · asked as “justify that systems that think/act rationally are part of AI” and as “define AI from the dimension of behavioural process and thought process”

	Humanly (measured against people)	Rationally (measured against an ideal)
Thinking	Thinking humanly – the cognitive modelling approach. Build a precise theory of how the human mind works, then write it as a program. Needs psychology and introspection. Eg General Problem Solver, Newell and Simon, 1961. Criticism: we have no scientific theory of the brain's internal activity, so the model cannot be verified.	Thinking rationally – the laws of thought approach. Encode correct reasoning as formal logic, then let an inference engine derive conclusions. Eg “Socrates is a man; all men are mortal; therefore Socrates is mortal.” Criticism: informal, uncertain knowledge is very hard to state in logical notation, and solving a problem in principle is not solving it in practice.
Acting	Acting humanly – the Turing test approach. Model what people do, not how they think. Judged purely on external behaviour. Eg ELIZA. This is the box §1.2 lives in.	Acting rationally – the rational agent approach. Do whatever is expected to achieve the best outcome. Covers reflex action, which needs no reasoning at all. The approach modern AI actually uses, because it is general and it can be measured. This is the box §1.3 lives in.

How to answer “justify that X is part of AI”

- Name the box, give its yardstick, give an example system, give its criticism.
- **Thinking rationally** is part of AI because logic-based inference (FOPL, resolution, Ch 4) derives new true facts from known ones without any human in the loop.
- **Acting rationally** is part of AI because an agent that maximises its performance measure in an uncertain environment is doing what intelligence is *for*, whether or not it reasons the way people do.
- **Say which is preferred and why:** acting rationally, because rationality is mathematically well defined and generalises, while “humanly” needs a theory of mind we do not have.

Types of AI by capability

- **Artificial Narrow Intelligence (ANI, weak AI):** one task or one narrow domain. Everything that exists today. Eg spam filter, chess engine, a speech recogniser.
- **Artificial General Intelligence (AGI, strong AI):** any intellectual task a human can do, transferring skill between domains. Does not exist.
- **Artificial Super Intelligence (ASI):** exceeds human ability in every field including creativity and social skill. Hypothetical.

Weak Vs strong AI, the philosophical sense

- **Weak:** the machine *simulates* thinking.
- **Strong:** the machine genuinely *has* a mind.
- Searle's Chinese Room argues strong AI is impossible: symbol manipulation is not understanding. verified/added

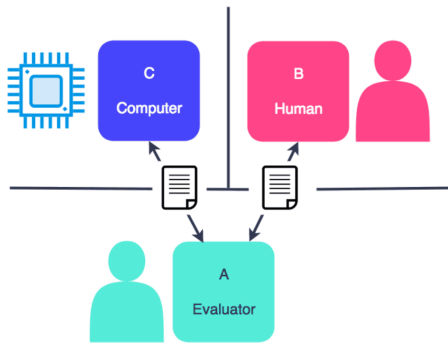
1.2 The Turing test

What is the Turing test? When is a machine said to have passed it? 2 4 5 7

TOP 10/41 70 Bh 5, 69 Po 7, 75 Ba 2, 77 Ch 3, 78 Ba 2, 79 Ash 4, 79 Bh 5, 81 Ba 6, 77 Ch 4, 76 Bh 3 · the single most-asked question in the subject's first chapter

The setup

- Proposed by **Alan Turing**, 1950, in *Computing Machinery and Intelligence*, to replace the unanswerable “can machines think?” with a testable question, “can machines behave intelligently?”
- Three rooms, no sight or sound between them:
 - a human **interrogator**,
 - a human **respondent**,
 - the **machine**.
- The interrogator questions both over a text-only channel (a teletype, so voice and handwriting give nothing away) and must say which is which.



Draw this: the arrangement is the definition. A is the interrogator, B the human respondent, C the machine. The wall and the text-only channel are the two things that must appear – they are what stops the test measuring anything except behaviour. A knows one of B, C is a machine and must say which.

When it is passed

- The machine passes when the interrogator **cannot do better than chance**, i.e. is wrong about half the time.
- Turing’s own operational figure: an average interrogator has **under a 70 % chance** of identifying correctly after **5 minutes** of questioning, so the machine fools them **30 %** of the time.
- **Write both:** the half-the-time principle, then the 5-minute / 30 % figure as the concrete criterion.

The four capabilities a machine needs to pass

- **Natural Language Processing** → to communicate in English at all.
- **Knowledge Representation** → to store what it knows and what it is told during the interview.
- **Automated Reasoning** → to answer questions and draw new conclusions from what is stored.
- **Machine Learning** → to adapt to new circumstances and detect patterns it was not given.

Total Turing test adds two more

- The plain test deliberately avoids physical contact, so it never tests perception or manipulation. The **total Turing test** puts a video link and a hatch for passing objects into the setup, adding:
 - **Computer vision** → to perceive objects,
 - **Robotics** → to manipulate them and move about.
- **6 capabilities in total.** (77 Ch and 76 Bh ask for the total test by name)

ELIZA, the standard example

- Written at MIT, **1964–66**, by **Joseph Weizenbaum**. First script **DOCTOR**, imitating a Rogerian psychotherapist.
- Mechanism: a table of **syntactic patterns**, each with a canned reply that echoes part of the input back after simple substitutions (*my* → *your*). **No understanding, no knowledge base, no memory.**
- Weizenbaum was disturbed by the reaction: psychiatrists thought it clinically useful and users confided in it.
- **The exam point:** ELIZA shows a machine can be *taken* for a human without any intelligence behind it, which is the standing objection to the test itself.

Importance of the Turing test

- Turns a philosophical question into an **operational, repeatable** one.
- Sets a **behavioural** yardstick: judge output, not internal mechanism, so it does not need a theory of mind.
- Defines the **research agenda**: its four required capabilities are four whole subfields of AI, and this syllabus is mostly those four.
- Still the benchmark by which conversational systems are reported.

Limitations

- Rewards **deception**, not competence. Passing needs human-like error and delay, so a machine that answers arithmetic instantly and correctly *fails*.
- **Not reproducible** and not mathematically definable; the result depends on the interrogator.
- Tests only what people happen to do, so it caps machine intelligence at human behaviour rather than measuring intelligence itself.

If the Turing test is passed, does that show computers exhibit intelligence? 2 3 4 7

HOT 4/41 72 Ma 7, 73 Bh 7, 81 Bh 4+3, 80 Ch 2+4 · **an argue-both-sides question**: take a side, but the marks are in giving both cases first. 81 Bh and 80 Ch tack the *reverse* Turing test on the end

Case for “yes”

- Intelligence has **no agreed internal definition**, so behaviour is the only thing that can be measured at all.
- We attribute intelligence to *other people* on exactly this evidence, and nothing else. Refusing the machine the same inference is inconsistent.
- Passing is not a trick: it requires NLP, knowledge, reasoning *and* learning working together, which is a real and general capability.

Case for “no” (the stronger answer)

- **Imitation is not cognition.** ELIZA fooled people with pattern substitution and knew nothing.
- **Searle’s Chinese Room**: a person following symbol-manipulation rules can answer Chinese questions perfectly while understanding no Chinese. Syntax does not give semantics. **verified/added**
- Tests only **one narrow channel**, text conversation. Says nothing about perception, physical skill, creativity or goals.
- **Recommended stance**: passing shows *intelligent behaviour*, not *intelligence*. It is a sufficient test of appearing intelligent and a necessary-but-not-sufficient test of being so.

Reverse Turing test 3 4

verified/added 81 Bh 3, 80 Ch 4 · 2 papers, both under 6 marks, so no tier chip · **not in any lecture deck in this folder**, researched and added, and it has been asked twice in the last three years

What it is

- The roles are swapped: **a machine interrogates, and a human must prove it is human.**
- The test succeeds when the machine can tell human from machine, which is the opposite goal to Turing’s.
- **CAPTCHA** is the everyday instance: *Completely Automated Public Turing test to tell Computers and Humans Apart*. A distorted-text or image-grid challenge that people pass easily and programs do not.

Turing Vs reverse Turing, side by side

- **Who judges**: human, Vs machine.
- **Who is on trial**: machine, Vs human.
- **Passed when**: the judge cannot tell them apart, Vs the judge *can*.
- **Purpose**: measure machine intelligence, Vs block bots from a web service.
- **Note the irony worth a mark**: every CAPTCHA solved by a program is a small Turing test passed, which is why CAPTCHAs keep getting harder.

1.3 Intelligent agents

What is an intelligent agent? Explain the different types with examples 1 2 6 8

HOT 7/41 74 Bh 1+6, 80 Bh 2+6, 80 Ash 2+6, 79 Jth 1+6, 75 Bh 8, 76 Ba 8, 71 Bh 4+4 · 75 Bh asks specifically **how a learning agent works**, so do not stop at the four standard types

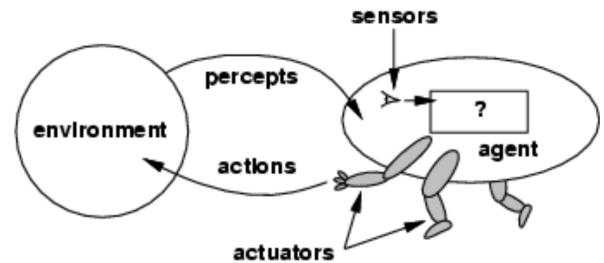
Definitions to write first

- **Agent:** anything that perceives its environment through **sensors** and acts upon that environment through **actuators**.
 - **Human agent:** eyes and ears as sensors; hands, legs and mouth as actuators.
 - **Robotic agent:** cameras and infrared range finders as sensors; motors as actuators.
 - **Software agent:** keystrokes and file contents as sensors; screen writes and network packets as actuators.
- **Percept:** what the agent perceives at one instant. **Percept sequence:** everything it has ever perceived.
- **Agent function** maps percept sequence to action, $f : P^* \rightarrow A$.
- **Agent program** is the concrete implementation that runs on the hardware.
- **Agent = architecture + program.** One line, and it is often a mark on its own.

Rational agent

- For each possible percept sequence, a rational agent selects the action expected to **maximise its performance measure**, given the evidence in the percept sequence and whatever built-in knowledge it has.
- **Performance measure:** the objective criterion for success, fixed by the designer, not the agent. Eg for a vacuum cleaner: dirt collected, time taken, electricity used, noise made.
- **Rationality is not omniscience.** An omniscient agent knows the actual outcome of its actions; that is impossible. Rationality maximises *expected* performance given what is knowable.
- **Autonomy:** an agent is autonomous when its behaviour is determined by its own percepts and experience rather than only by built-in knowledge. Autonomy requires learning.
- **Information gathering** and **exploration** are rational acts: it can be rational to act now purely to get a better percept later.

The five agent types, in increasing generality



The agent, its sensors and actuators, and the environment loop.

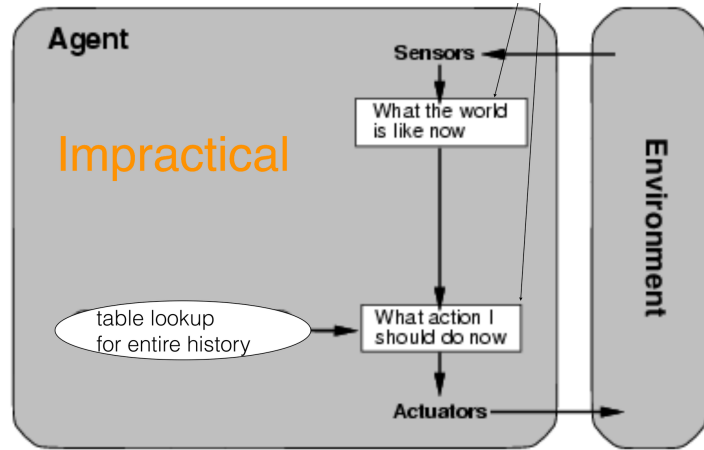
Draw this first in any agent answer; every later diagram is this one with a box added inside the agent.

Environment properties (worth 2 marks when asked)

- **Fully Vs partially observable:** can the sensors see the whole relevant state?
- **Deterministic Vs stochastic:** does the next state follow from the current state and action alone?
- **Episodic Vs sequential:** does the current action affect later ones?
- **Static Vs dynamic:** does the world change while the agent deliberates?
- **Discrete Vs continuous:** countable states and actions, or not?
- **Single agent Vs multi-agent:** is anything else optimising against it?
- Eg a taxi driver's world is partially observable, stochastic, sequential, dynamic, continuous and multi-agent. **The hardest kind on every axis.**

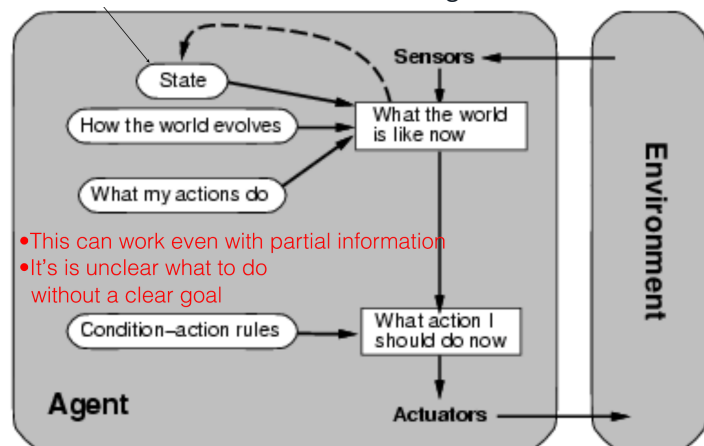
1. Table-driven agent

- Holds a lookup table indexed by the *entire percept history*, and reads the next action out of it.
- **Impractical**: the table grows exponentially with the length of the history. A taxi with a camera would need more entries than there are atoms in the universe.
- Included only to show why the later types exist.



3. Model-based reflex agent

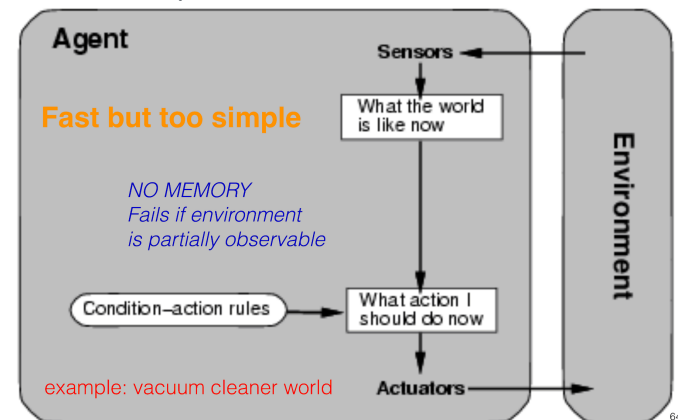
- Keeps **internal state** = the part of the world it cannot currently see.
- Needs two pieces of knowledge, together called the **model of the world**:
 - how the world evolves on its own,
 - what my actions do to it.
- **Fixes partial observability**. Eg a taxi remembering which cars are in the lanes it cannot see right now.



- This can work even with partial information
- It's unclear what to do without a clear goal

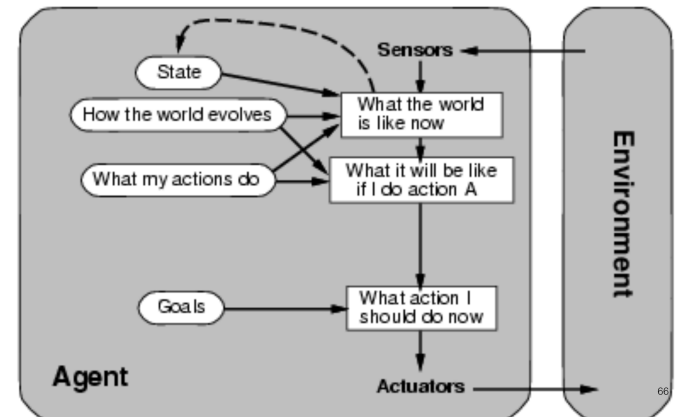
2. Simple reflex agent

- Acts on the **current percept only**, through **condition-action (if-then) rules**. Implemented as a production system (Ch 2).
- **Stateless**: no memory of past world states.
- Eg
 - if car-in-front-is-braking then initiate-braking;
 - the vacuum-cleaner agent that sucks whenever the current square is dirty.
- **Fails when the environment is partially observable**, and can loop forever in that case.



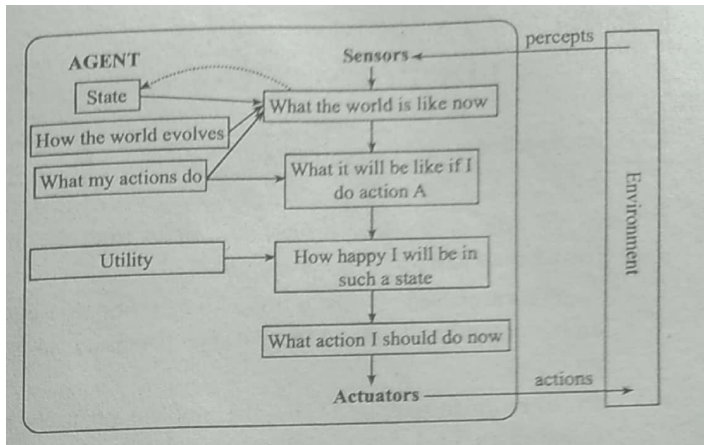
4. Goal-based agent

- Adds **goal** information describing desirable situations, and asks “what will it be like if I do action A?”
- Considers the **future**, so it can plan and search (Ch 3), not just react.
- **Far more flexible**: change the goal and the behaviour changes, with no rules rewritten. A reflex agent needs every rule rewritten.
- Eg a taxi told to go to a different destination.



5. Utility-based agent

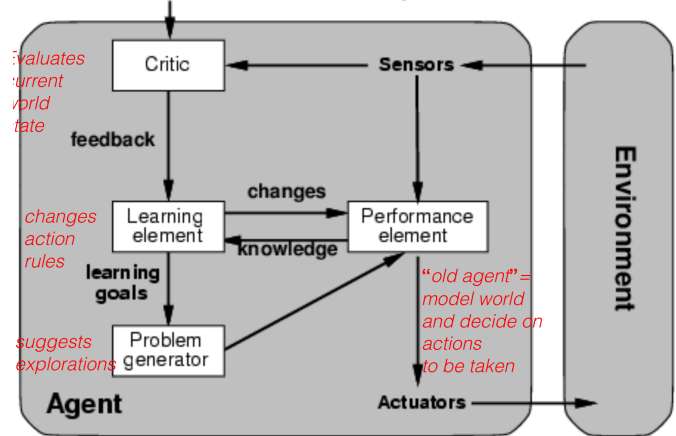
- A goal is binary: reached or not. A **utility function** maps a state to a number saying **how happy** the agent is there.
- Needed for two cases a goal cannot settle:
 - **conflicting goals** – fast Vs safe Vs cheap: utility trades them off;
 - **uncertain goals** – pick the action with the highest *expected* utility.
- Eg a taxi choosing between a quick risky route and a slow safe one.



Learning agent (asked by name, 75 Bh 8)

- Any of the four above, wrapped in machinery that **improves it with experience**. Four parts:
 - **Performance element** – the old agent: takes percepts, chooses actions.
 - **Critic** – judges the outcome against a fixed **performance standard** and reports feedback. The standard must be external, or the agent could just lower its own bar.
 - **Learning element** – uses that feedback to change the performance element.
 - **Problem generator** – deliberately suggests **exploratory** actions that look suboptimal now, so the agent discovers better behaviour later.
- **The exam point:** learning is what makes an agent truly autonomous, which is the link straight to §1.9 and to Ch 6.

Performance standard *better modeling, new action rules, etc.*



1.4 PEAS: specifying the task environment

Define PEAS. Give the PEAS information for a named system 2 3 5 6

HOT 7/41 80 Ba (medical diagnosis) 3, 78 Ch 4, 74 Bh 3, 81 Ash (taxi + robot) 5, 82 Ka (robot) 3, 82 Ba (self-driving car) 6, 82 Bh (medicine-delivery drone) 5 · **asked with a different system every single time**, so learn the four rows, not four memorised answers

What and why

- Before designing an agent, its **task environment** must be specified. PEAS is that specification:
 - P – Performance measure:** what counts as success. Must be about the *environment*, not the agent's internal state.
 - E – Environment:** what the agent operates in and on.
 - A – Actuators:** how it acts.
 - S – Sensors:** how it perceives.
- Why it must come first:** the performance measure decides what “rational” means, and the sensors and actuators decide what the agent can even represent. Design before PEAS is design without a specification.

Three mistakes that cost marks

- Writing the performance measure as “*work well*”. Give measurable terms: safe, fast, legal, comfortable, maximise profit.
- Putting an actuator in the sensor row. A horn is an actuator; a microphone is a sensor.
- Forgetting that a **display** is an actuator when the agent uses it to tell a person something.

Self-driving car (82 Ba 6)

- P:** reach the destination safely, obey traffic law, minimise journey time and fuel, ride comfort, zero collisions.
- E:** roads and lane markings, other vehicles, pedestrians and cyclists, traffic signals, road signs, weather, the passenger.
- A:** steering, throttle, brakes, gear selection, indicators, horn, headlights, passenger display and speaker.
- S:** cameras, LIDAR, radar, ultrasonic parking sensors, GPS, IMU and accelerometer, wheel-speed sensors, microphone.

Ready-made answers for the systems actually asked

- Taxi driver** (81 Ash)
 - P:** safe, fast, legal, comfortable trip, maximise profits.
 - E:** roads, other traffic, pedestrians, customers, weather.
 - A:** steering, accelerator, brake, indicator, horn, display.
 - S:** cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard.
- Medical diagnosis system** (80 Ba)
 - P:** healthy patient, minimise costs and lawsuits.
 - E:** patient, hospital, staff.
 - A:** screen display of questions, tests, diagnoses, treatments, referrals.
 - S:** keyboard entry of symptoms, findings and the patient's answers.
- Part-picking robot in a plant** (81 Ash, 82 Ka)
 - P:** percentage of parts placed in the correct bin.
 - E:** conveyor belt with parts, the bins.
 - A:** jointed arm, gripper.
 - S:** camera, joint angle sensors.

Medicine-delivery drone (82 Bh 5) verified/added

- No source note covers this one;** state your assumptions, as the paper explicitly allows, then give four clean rows.
- P:** medicine delivered intact, to the right address, within the time window, maximum payload per charge, no collision, no airspace violation.
- E:** air corridor, buildings, terrain and hills, trees and power lines, weather and wind, other drones and aircraft, the recipient, the dispatch centre.
- A:** rotors and motors, payload release hatch, landing gear, status lights, radio downlink to the dispatch centre.
- S:** GPS, camera, LIDAR or ultrasonic altimeter, IMU and gyroscope, barometer, wind and temperature sensors, battery gauge, radio uplink.

1.5 Importance and limitations of AI; AI Vs human intelligence

Differentiate natural and artificial intelligence. Can AI replace humans? 3 4 5 7

PIN 3/41 79 Ch 1+3+4, 78 Po 3+5, 73 Bh 7 · growth areas asked at 71 Bh 4, disadvantages at 78 Ba 2

	Human intelligence	Artificial intelligence
Origin	Biological, inherited and grown	Engineered, programmed and trained
Learns from	A few examples, plus intuition and context	Very large data sets, no context of its own
Speed	Slow, tires, needs rest	Constant, no fatigue, runs continuously
Accuracy	Errs when tired or emotional	Exact and repeatable within its domain
Breadth	General: transfers skill to a new task	Narrow: fails outside the trained task
Creativity	Original ideas, imagination	Recombines what it was trained on
Emotion	Empathy, judgement, ethics	None; only what is coded as a measure
Adapting	Handles the wholly unfamiliar	Needs retraining or reprogramming
Cost	Salary, training time	High to build, near-zero to copy

Importance and advantages of AI

- **Automates repetitive work** → people move to judgement work.
- **Better decisions:** finds patterns in data volumes no person can read.
- **Available always:** no fatigue, no shift limits, consistent output.
- **Works where people cannot:** deep sea, space, reactor cores, disaster zones, mine clearance.
- **Fewer human errors** in exact, repetitive tasks.
- **Reaches scarce expertise:** an expert system puts specialist knowledge in a village clinic.
- **Speeds research:** drug discovery, protein folding, materials.

Growth areas (71 Bh)

- Machine learning and deep learning · NLP and speech · computer vision · robotics and autonomous vehicles · expert and decision-support systems · recommender systems · medical imaging and diagnosis.

Can AI replace humans? Answer: in tasks, yes; in roles, no

- **Where it replaces:** repetitive, rule-bound, data-heavy work. Data entry, first-line support, defect inspection, routine diagnosis, translation drafts.
- **Where it cannot, and why:**
 - **No common sense** outside the training distribution.
 - **No moral or legal responsibility;** someone must be accountable.
 - **No genuine creativity or empathy,** both needed in teaching, care, negotiation, design.
 - **Depends on data** that people must collect, label and correct.
- **Conclusion to write:** AI is a tool that shifts human work up the skill ladder, not a replacement for the worker. The realistic risk is *displacement of specific tasks*, not obsolescence of people.

Disadvantages and limitations (78 Ba 2)

- **Cost:** high to build, and needs specialists to maintain.
- **No common sense:** fails absurdly on inputs slightly outside its training.
- **No creativity:** recombines, does not originate.
- **Job displacement** in routine occupations.
- **Opaque:** a neural network cannot explain its decision, which is unacceptable in medicine and law.
- **Inherits bias** from its training data, then applies it at scale.
- **No ethics or accountability,** so misuse is easy: surveillance, autonomous weapons, deepfakes.
- **Data-hungry and energy-hungry.**

Two named ethical positions worth a line

- **Weizenbaum (1976):** a real AI would be autonomous and would not share our motives or ethics, so the research is unethical.
- **Asimov's three laws of robotics:** do not harm a human or by inaction allow harm; obey humans; protect itself. Precedence runs lower number first.

1.6 AI and related fields

Describe AI and its subfields; which fields contributed to AI? 4

76 Bh 4 · asked once, but it is a two-minute answer and it recurs as a sub-part of "define AI"

What each field gave AI

- **Philosophy:** logic, the idea of the mind as a physical system, foundations of learning, language and rationality.
- **Mathematics:** formal representation and proof, algorithms, computation, decidability and intractability, probability, statistics, linear algebra.
- **Psychology:** perception, motor control, adaptation, and the experimental methods behind cognitive modelling.
- **Economics:** formal theory of rational decision, utility, game theory.
 - This is where the *utility* in a utility-based agent comes from.
- **Linguistics:** grammar, syntax and knowledge representation. NLP sits directly on it.
- **Neuroscience:** the physical substrate of mental activity. Neural networks are a deliberate, loose imitation of it.
- **Control theory:** stability, feedback, optimal agent design. The agent-environment loop is a control loop.
- **Computer engineering:** the machines fast enough to run any of it.

Fields AI overlaps with, and how it differs

- **Machine learning** is a subfield of AI, not a synonym: AI also covers search, logic and planning, which do not learn at all.
- **Data mining** (Ch 6 and the Data Mining elective) shares algorithms but aims at *discovering patterns in stored data*; AI aims at *acting well*.
- **Robotics** supplies the body; AI supplies the decision making.
- **Cognitive science** is the joint enterprise of AI and psychology: AI wants machines that work, cognitive science wants a theory of the mind.
- **Operations research** shares optimisation and search but assumes a fully specified model.

Programming languages of AI

- **LISP:** McCarthy, late 1950s. Functional, symbolic, interactive.
- **PROLOG:** France 1970, developed at Edinburgh 1975–79. Logic programming: a program is facts + rules, and running it is inference. **The syllabus lab is in one of these two.**

1.7 Brief history of AI

Discuss the brief history of AI with chronological development 2 5 6 7

PIN 3/41 70 Ma 2+6, 71 Bh 2+5, 72 Ma 7 · 72 Ma asks specifically **which period was the “AI research failure” period and why**, so learn the two AI winters, not just the timeline

The founding era, 1943–1979

Year	Event
1943	McCulloch and Pitts : Boolean circuit model of the neuron. The first neural network.
1950	Turing , <i>Computing Machinery and Intelligence</i> . The Turing test.
1956	Dartmouth meeting : the term “Artificial Intelligence” is adopted. The birth of the field.
1950s	Samuel’s checkers program, which learned and beat its author; Newell and Simon’s Logic Theorist; Gelernter’s Geometry Theorem Prover (1959).
1957	Rosenblatt’s Perceptron.
1958	McCarthy defines LISP and describes the Advice Taker.
1963	McCarthy starts the Stanford AI lab. Slagle’s SAINT solves first-year calculus.
1965	Robinson’s resolution : a complete theorem-proving algorithm for first-order logic. Ch4 is built on it.
1966	ELIZA , Weizenbaum.
1969–79	Knowledge-based and expert systems : DENDRAL , MYCIN , SHRDLU . Minsky’s frames (1975).

The “AI research failure” period

- Two collapses in funding and confidence, both called **AI winters**. The question almost always means the first.
- **First AI winter, 1966–73**, the “dose of reality”. In 1957 Simon predicted a computer chess champion and a major theorem proved within **10 years**; both took roughly **40**. Three reasons it failed:
 - **No subject knowledge**. Early programs succeeded by *syntactic manipulation* alone. Machine translation is the standard example: word-by-word substitution with no meaning behind it.
 - **Intractability**. Programs solved toy problems by trying combinations of steps. Scaling the problem exploded the search, and the 1965 discovery of NP-completeness explained why faster hardware could not fix it.
 - **Structural limits**. Minsky and Papert’s *Perceptrons* (1969) proved a single-layer perceptron cannot represent XOR, and neural network research nearly stopped for fifteen years. **This is the same XOR result you prove in Ch 7.**

Industry, science and the modern era, 1980–

Year	Event
1980	AI becomes an industry . XCON saves DEC an estimated \$40 million a year.
1986	Neural networks return , on back-propagation.
1987	AI becomes a science : results measured against benchmarks, and statistics and probability adopted.
1995	The intelligent agent becomes the unifying idea of the field.
1997	Deep Blue beats Kasparov at chess.
2011	IBM Watson wins <i>Jeopardy!</i>
2016	AlphaGo beats Lee Sedol at Go.
2020–	Large language models, GPT-3 onward.

- **Write it as a table, not as prose.** Year, then event, then one clause on why it mattered. Dates with no significance score badly.

The second winter, and what ended both

- **Second AI winter, late 1980s to early 1990s**: the expert-system market collapsed. The systems were **brittle** outside their narrow domain, **could not learn**, and cost more to maintain than they saved.
- **What ended each**:
 - the first ended with **knowledge-based systems** – stop searching blindly and put the domain knowledge in;
 - the second ended with **statistical machine learning**, then cheap parallel hardware and large data sets.
- **The pattern behind both**, and the sentence that earns the last mark: each winter followed a period of **over-promising on a technique that did not scale**, and each ended when the field adopted a *different* technique rather than more of the same.

1.8 Applications of AI

Discuss fields of daily life where AI has been applied 2 4 6 7

TOP 8/41 69 Bh 7, 72 Ash 7, 73 Ma 6, 81 Ch 2+4, 75 Ba, 77 Ch, 78 Ba, 78 Po · usually the tail of a Turing-test or define-AI question, which is why it is worth 4 marks and two minutes. Most papers ask for **any two**, explained

How to answer for full marks

- The question says *explain two fields*, not *list ten*. A list scores badly.
- For each field give: **the task**, **the AI technique behind it**, and **a named example**. Three sentences each is the whole answer.
- 81 Ch asks for applications **as a problem-solving tool**, so frame each one as “the problem, and how AI solves it”.

Two safe choices, worked

- **Healthcare**. Problem: expert diagnosis is scarce and slow. Technique: neural networks over medical images (Ch 7), plus rule-based expert systems for diagnosis. Examples: tumour detection on CT and MRI, IBM Watson for Oncology, diabetic-retinopathy screening from retinal photographs.
- **Transport**. Problem: driving needs continuous perception and decision under uncertainty. Technique: computer vision and sensor fusion feeding a utility-based agent, plus A* and its variants for route search (Ch 3). Examples: self-driving cars, Google Maps ETA prediction, Pathao and inDrive dispatch and pricing.

The wider list, grouped

- **Everyday software**: search-engine ranking and autocomplete, social-media feeds, Netflix and YouTube recommendation, spam filters, photo tagging.
- **Language**: machine translation, speech recognition, chatbots and virtual assistants, Nepali NLP tools (Ch 7).
- **Finance**: fraud and money-laundering detection, credit scoring, algorithmic trading.
- **Industry and agriculture**: robotic assembly and part picking, predictive maintenance, visual quality inspection, crop-disease detection from leaf images, yield prediction.
- **Games**: Deep Blue, AlphaGo, and the bots in Dota 2 and PUBG.
- **Science and space**: astronomical object classification, protein folding, space-mission monitoring, drug discovery.
- **Public services**: weather and flood forecasting, traffic signal control, disaster response, biometric identification.

Nepal-specific examples, if the question invites them

- Pathao and inDrive dispatch · eSewa and Khalti fraud detection · Nepali speech and OCR · landslide and flood early warning · crop-disease apps.

1.9 Knowledge and learning

What is knowledge and learning? Why are they important? 2 4 5 8

HOT 5/41 73 Bh 4+4 , 74 Ma 2+2+2+4 , 72 Ash 2+2+2+4 , 74 Bh 4+4 , 79 Jth 4+4 · also inside the Turing question at 79 Ash 2+2 and 71 Ma 2
 · the quotable line the examiner wants back: "Learning is an essential characteristic for intelligent agents"

	Data	Information	Knowledge
What	Raw facts, unorganised	Data processed into a meaningful, usable form	Information organised and understood, ready to act on
Form	Numbers, symbols, a database	Tables, reports, summaries	Rules, procedures, models
Answers	nothing on its own	who, what, where, when	how and why
Example	$V = 12, I = 6, R = 2$	$V = I \times R$ holds for these readings	Use $V = IR$ to design a circuit

- **Intelligence** sits one step above knowledge: the ability to use knowledge to solve new problems and adapt to new situations.
- **Write the chain:** data → information → knowledge → intelligence. Each step adds structure and context.

Knowledge, and why AI needs it

- **Knowledge:** the condition of knowing something with familiarity gained through experience or association; what lets us explore and decide.
- Stored by people in **natural language**, by machines as **symbols, strings and structures**.
- Forms it takes: simple facts, relations between concepts, **associations, inheritance hierarchies**, procedures.
- **Importance:** enough knowledge is what produces intelligence. A search algorithm with no domain knowledge is blind (Ch 3); the same algorithm with a heuristic is informed and vastly faster.
Knowledge is what turns search into intelligence.
- **The three designer problems** (they run through Ch 4 and Ch 5):
 - **Knowledge acquisition** – getting it out of the domain and its experts.
 - **Knowledge representation** – encoding it: logic, semantic nets, frames, rules.
 - **Knowledge manipulation** – inferring new conclusions from it.

Learning

- **Herbert Simon (1983):** "Learning is the phenomenon of knowledge acquisition in the absence of explicit programming."
- Fuller form: learning denotes **changes in the system that are adaptive**, in the sense that they let it do the same task, or a task drawn from the same population, more efficiently and more effectively next time.
- **Three factors in any learning system:**
 - **Changes** – what in the system is altered, and how it is represented.
 - **Generalisation** – performance must improve on *similar* tasks, not only the one seen. Without this it is memorisation.
 - **Improvement** – the change must not degrade performance; the system must detect and prevent that.
- **Knowledge Vs learning in one line:** knowledge is the *content* an agent holds; learning is the *process* that changes that content.

"Learning is an essential characteristic for intelligent agents"

- **Agree, and justify in four steps:**
 - **The designer cannot foresee every situation.** A fixed rule table is only as good as its author's imagination, and the table-driven agent shows how quickly enumeration becomes impossible.
 - **Environments change.** Roads are rebuilt, spam changes its wording, prices move. A non-learning agent's performance decays with no fault of its own.
 - **Autonomy requires it.** An agent whose behaviour comes only from built-in knowledge is running the designer's intelligence, not its own. Learning is exactly what makes it autonomous.
 - **The Turing test demands it.** Machine learning is one of the four capabilities needed to pass, so an agent that cannot learn cannot be judged intelligent on the field's own benchmark.
- **Example to close with:** a spam filter shipped with fixed rules is beaten within weeks; one that learns from each user's marking of messages keeps working. Same architecture, and learning is the only difference.

Belief, hypothesis and knowledge 4

71 Bh 4 · asked once, as the tail of the growth-areas question. A three-row table is the whole answer

Term	Means	Confidence
Belief	Any statement an agent holds to be true. It need not be verified and may turn out false.	Subjective, unverified.
Hypothesis	A belief <i>put forward for testing</i> , backed by some evidence and held provisionally until confirmed or refuted.	Justified belief, not yet confirmed.
Knowledge	A belief that is <i>true</i> and <i>justified</i> . Verified, and safe to reason from.	Confirmed.

- **The relation, in one line:** knowledge $\subset$ hypothesis $\subset$ belief. Every hypothesis is a belief; only a confirmed, true one becomes knowledge.
- **Why AI cares:** an agent must act on beliefs, because it rarely has knowledge. That is precisely why Ch 4 needs probability and belief networks rather than plain logic.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network

Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
MOP	memory organization packet	isa	the class-to-class link of a semantic net (<i>instance</i> is the individual-to-class one)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = **CT 653** (BCT, III-II, 26 papers) · `monospace` = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. `syllabus, no PYQ yet` marks a syllabus topic never yet examined but written up anyway. `verified/added` marks anything researched and added that no source note in this folder covers.